

TECHNOLOGY OF SODIUM COOLED FAST REACTOR FUELS IN THE U.S.

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1. Introduction

The sodium cooled fast reactor fuel development programme in the United States is currently concentrating on three fuels: uranium-plutonium compounds of oxides, carbides, and alloys, in that order of priority. Although not discussed in this paper, other fuel types such as molten plutonium alloys, nitrides, and combinations of sulphides, phosphides, carbides, and nitrides are being investigated to a lesser extent.

The mixed oxides are favoured by the U.S. Atomic Energy Commission, U.S. industry and major foreign programmes, as the current choice for the prototype reactor fuel. Irradiations have been performed to approximately 9 atom per cent (at.%) burnup in a thermal flux in the U.S. and exceeding 5 at.% burnups on a relatively large number of specimens in Dounreay.

The mixed carbides, when compared to the mixed oxides, offer a potential improvement in breeding through their higher metal atom density and higher fuel performance (as a result of the higher thermal conductivity). Irradiations in excess of 10 at.% in a thermal flux in the U.S. and in excess of 5 at.% in Dounreay have been performed; the results justify continued interest in the mixed carbides. Fuel swelling, fuel to clad compatibility, and control of fuel composition (stoichiometry) are the major problem areas to be investigated.

The metal alloy fuels have a higher metal density, thermal conductivity, temperature induced coefficient of expansion and breeding gain potential when compared to the ceramics. Fuel to clad compatibility, fuel swelling, and low melting temperatures have been the major problems encountered with uranium-plutonium-fissium alloys. Recent investigations with the

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U-Pu-Zr ternary alloys have indicated improvement to the three aforementioned problem areas. The fuel is compatible with 304-SS up to 800°C and burnups in excess of 5 at.% have been achieved.

Satisfactory performance of all the above fuels may be realized if cladding integrity can be assured. The cladding materials being considered include austenitic stainless steels, intermediate nickel alloys and refractory alloys. The austenitic steels are compatible with sodium and considerable commercial experience exists, but these alloys suffer from neutron induced embrittlement and low strength at anticipated fast reactor operating conditions. The intermediate nickel alloys have better strength characteristics in the 500–700°C temperature range when compared to the austenitic steels, but encounter severe reductions in ductility due to a combination of neutron induced damage and thermal effects. Further, compatibility with fuel and coolant are not as well known and understood at this time when compared to austenitic alloys, such as 304-SS. The refractory alloys offer good strength characteristics and, potentially, neutron damage resistance; however, coolant compatibility, fuel compatibility, fabrication costs, and core neutronics (in some cases) are current problems to be considered.

Irradiation testing of the candidate fuels and materials for the sodium cooled fast breeder reactors is now carried out in a number of thermal reactors in the U.S. and in EBR-II, located at the National Reactor Testing Station in Idaho. To increase the U.S. testing capacity, plans are now under way to use the Enrico Fermi reactor. However, for the long term testing requirements, a test reactor capable of achieving high fast neutron fluxes for accelerated and advanced fuels and materials testing will be required. To meet this requirement, a fast flux test facility (FFTF), which will contain closed loops, is being designed. It is intended to initiate testing in the FFTF during late 1972 or early 1973.

2. Mixed Oxides

The urania-plutonia mixed oxide fuels have been under development in the U.S. since 1959. The emphasis on mixed oxides is primarily because of (a) the relative ease and economy of fabrication, (b) its demonstrated ability to withstand exposures to 99,000 MWd/t,⁽¹⁾ (c) the wealth of UO₂ know-how and experience available within the U.S., and (d) industrial interest.

Synthesis and fabrication processes

Homogeneity

Perhaps the most important question in the fabrication of mixed oxide fuels concerns the degree of non-homogeneity that is acceptable for a given reactor case.⁽²⁾ This question of solid solution (U,Pu)O₂ versus physical

blends of PuO_2 in UO_2 has been generally deferred in the USAEC programme. A long range goal, however, must be to obtain adequate homogeneity by the more economical processes involving some type of physical blending of UO_2 and PuO_2 .^{*} There is evidence that entirely adequate comminution and blending processes are at hand to prepare homogeneous fuels from physical blends.⁽³⁾ The question of quality assurance then becomes paramount, requiring that methods⁽⁴⁾ be developed to measure and evaluate degrees of bulk homogeneity and particulate sizes.

Stoichiometry

Conventional oxide fuels for present economic power reactors typically employ essentially stoichiometric fuel, UO_{2+x} , $0 \geq x \leq 0.02$. This has proven satisfactory for relatively low burnup applications (10,000 to 25,000 MWd/t) in which stoichiometry changes with burnup are small. Fast reactor fuels, of the type $(\text{U,Pu})\text{O}_2$, are required to attain average burnups of 75,000 to 100,000 MWd/t to be of economic interest, and local peak burnups of up to 130,000 MWd/t.

A recent summary and status of some fuel preparation and properties data and fuel performance observations which led to a tentative preference for an oxygen to metal ratio of 1.97 to 1.98 for some compositions of mixed oxide fuel were recently reported.^(5,8)

Density

The problem of fuel swelling at high exposures is obtaining ever-increasing attention. The question of how much void space is required, and how it should be deployed in the fuel material, has not been determined. As in other countries, there is no agreement concerning the best method of preparing low density pellets. One approach is to "deaden" the powder so that it will not sinter to a high density; the second, apparently more promising, approach is to pre-press at a high pressure, granulate, and press at a lower pressure prior to sintering.⁽⁶⁾ Annular pellets, low density solid pellets, and mechanically packed particles are being studied both out-of-reactor and in-reactor, including fast reactor irradiations in EBR-II.

3. Mixed Carbides

Uranium-plutonium carbide fuel has an excellent potential for producing a competitive fuel cycle cost among the three fuel systems considered in this paper: metal, oxide, and carbide. This is shown in a number of recent comparison studies and reviews.⁽¹⁵⁻²⁰⁾ Three properties highlight the attraction of carbides compared to oxides: thermal conductivity, fission gas retention,

^{*} It is noted that the French are proceeding with fabrication of the Rapsodie fuel by a physical blending route,⁷ with no substantial difficulties having been reported in meeting the close dimensional and compositional tolerances.

and fuel density. The thermal conductivity is four to ten times higher than that of uranium dioxide.

The fission gas retention by carbides is considerably better than that of oxides even at equivalent temperatures. Fission gas retention is feasible to burnups of 17×10^{20} fiss/cm³ (50,000 MWd/t).⁽²⁵⁾ The fissionable fuel density of carbides is 30–40% greater than that of oxides, although lower than that of metallic fuels. In common with oxides, the carbides have the advantages of high melting point and good dimensional stability over metallic fuels. The stability of the ceramics at high power and burnup permits the fuel cycle costs to be reduced almost proportionately to increasing burnup.

Fabrication Methods

Oxide and metal are the two most common raw materials for carbide preparation. Since recycling produces oxides several steps before it could produce metal, an economic incentive exists for investigating oxides as the starting material. The various carbide synthesis methods used in the U.S. have been: (a) carbon reduction of the fuel oxides, (b) hydrocarbon gas reaction with the fuel hydrides, and (c) melting metallic fuel with carbon, followed by casting.

Carbide synthesis

Carbothermic reduction of the mixed uranium–plutonium oxides has been studied at Argonne National Laboratory (ANL),⁽²¹⁾ Los Alamos Scientific Laboratory,⁽²²⁾ and United Nuclear Corporation (UNC)-Carborundum,⁽²³⁾ starting with mechanically mixed or solid solution oxides. The reaction is carried out between 1500°C and 1600°C in an inert atmosphere, or at a slightly lower temperature in vacuum. Considerable fabrication experience indicates that vaporization of plutonium in inert atmospheres is 1% or less. In vacuum, vaporization rates will be higher at an equivalent temperature, but the lower required reaction temperatures and times can keep the losses low.

The reaction of uranium and plutonium hydrides with hydrocarbon gas produces a good high purity carbide. A fluidized bed version of the process is being developed at ANL. The low reaction temperature (600–900°C) of the fluidized bed process is an advantage. Melting elemental constituents together has been used by most laboratories to produce experimental quantities of carbides. Large plutonium losses (through volatilization) occur in vacuum, but the losses can be reduced by melting under a partial pressure of an inert gas.

Carbide fabrication

Powders from any one of the synthesis processes can be used to make cold pressed and sintered shapes. This fabrication method has been investigated extensively.^(22, 23, 24) The use of nickel also reduces the stringency of the

particle size and powder surface purity (i.e. handling atmosphere purity) requirements necessary for high densities. Densities of 98% theoretical (average) are achieved by sintering at 1550°C with 0.1 to 0.2 weight per cent (wt.%) nickel sintering aid, and 92% theoretical by sintering at 1925°C without a sintering aid. The material without a sintering aid is single phase or has traces of dicarbide. The material with sintering aid always has a second phase of sesquicarbide.

The arc melting process has been used to synthesize material for sintering of pellets and to produce feed material for vibratory compaction at ANL. This process was used because homogeneous material with reproducible carbon composition and low oxygen content was desired. The optimum particle size for the pellets containing less than 1500 ppm oxygen was found to be approximately 40 μ . Desired densities of about 85% of theoretical were obtained after sintering at 1900°C for 2 hours. The microstructure of the pellets was single phase and chemical analyses showed that they contained from 4.64 to 4.72 wt.% C, 0.11 to 0.12 wt.% O, and 0.014 to 0.020 wt.% N.

Two vibratory-compacted U-Pu carbide systems are being studied at ANL—solid solution powder and mixed powders of UC and PuC. A group of fuel specimens of physically mixed UC-20 wt.% PuC powders have been made recently for capsule irradiation studies. Three size fractions of powders were blended and vibratory-compacted directly in the fuel cladding. The middle size fraction was PuC powder; the other two fractions were UC powder. In order to make up fuelled rods (~20 in. long) for irradiation in EBR-II, a basic study of the particle packing system provided the mathematical relationships necessary to predict the packing efficiencies in full length rods.⁽²⁶⁾ This procedure involves basically an infiltration technique in which fine powder is allowed to fill the interstices between larger particles which have first been vibrated into a compacted form.

Compatibility studies

The initial phase of the compatibility studies at ANL on (U,Pu)C consists of surveying the interaction of a number of cladding materials, including refractory metals and iron and nickel base alloys, with stoichiometric single phase (U,Pu)C over the temperature range of 650–1100°C. Brief comments on each of the systems studied to date are as follows: No reaction of single phase (U,Pu)C was noted with type 304-SS at 800°C for 42 days. At 1100°C, heavy grain boundary penetration was noted with eventual melting occurring between 17 and 42 days. With Hastelloy-X a reaction zone 8 μ thick was noted after 42 days at 800°C. With vanadium, no reaction was noted at 950°C up to 17 days but grain boundary penetrations were observed after 42 days. Tantalum, niobium, and molybdenum showed 4–8 μ reaction zones after 7 days at 1100°C. No reaction was found with chromium and tungsten under these same conditions.

TABLE 1. SUMMARY OF VIBRATORILY COMPACTED CARBIDE IRRADIATIONS

Specimen no.	Fuel	Jacket material	Wt. % C in PuC	Burnup		Maximum jacket surface temperature °C	Max. % Δ dia.	% ΔL	% ΔV	Gas release % theoretical
				at. %	Fiss./cm ³ $\times 10^{20}$					
F-7	PuC	304-SS	4.24	2.0	5.1	495	1.0	0.1	1.1	35.2
F-6	PuC	304-SS	6.22	5.4	13.1	625	0.8	Specimen melted	Jacket ruptured	25.7
F-8	PuC	304-SS	4.24	6.0*	16.0	675	14.3			
C-44	PuC	Nb-1Zr	3.40	2.2	5.3	625	0.5	Neg.	0.3	45.9
C-43	PuC	Nb-1Zr	4.24	5.3	14.0	550	0.1	Neg.	0.3	31
F-18	UC-20PuC	304-SS	6.42	1.4	3.4	505	0.3	Neg.	0.6	8.8
F-21	UC-20PuC	304-SS	3.40	1.4	3.5	430	0.5	Neg.	0.2	25.7
F-19	UC-20PuC	304-SS	6.22	5.8*	15.0	570	2.8	0.3	3.5	<1*
F-20	UC-20PuC	304-SS	4.24	6.8*	18.0	645	4.7	1.3	7.3	40
C-69	UC-20PuC	Nb-1Zr	6.42	1.2	3.1	405	0.5	Neg.	0.3	5.7
C-74	UC-20PuC	Nb-1Zr	3.40	1.6	4.1	455	0.3	Neg.	0.5	21.5
C-72	UC-20PuC	Nb-1Zr	6.22	6.8	18.0	635	0.2	Neg.	0.4	$\pm 1^*$
C-70	UC-20PuC	Nb-1Zr	4.24	6.8*	18.0	650	0.4	Neg.	0.5	20

* Based only on xenon release.

Irradiation studies

Early emphasis at ANL has been on irradiations of low density mixed carbides in the form of compacted physical mixtures of UC-20 wt.% PuC powders.⁽²⁷⁾ Specimens now being placed under irradiation, however, include solid solution pellets and powder fabricated as described in Section 2. A few vibratorily compacted PuC powder specimens have also been irradiated to obtain information on the single carbide system.

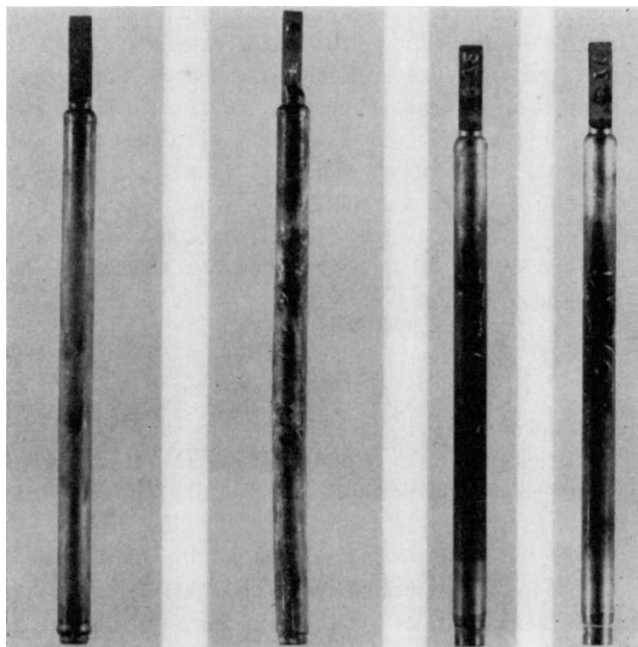


FIG. 1. Appearance of vibratorily compacted UC-20 wt.% PuC specimens after irradiation up to 6.8 at.% burnup. The two specimens on the left are jacketed in type 304 stainless steel; those on the right are jacketed in Nb-1 at.% Zr alloy.

The examination of four physically mixed UC-20 wt.% PuC and two PuC vibratorily compacted specimens irradiated in thin cladding to an estimated 5.3–6.8 at.% burnup has recently been completed except for metallography and burnup analyses. The results obtained so far are given in Table 1. All specimens had 0.0229 cm (0.009 in.) thick jackets and a fuel diameter of 0.396 cm (0.156 in.). The mixed carbide specimens are shown after irradiation in Fig. 1. Inspection of the data shows that the Nb alloy was highly effective in preventing dimensional changes, and that fission gas release was dependent on the carbon content of the PuC with hypostoichiometric fuel releasing considerably more gas. All mixed carbide specimens were intact. For the

PuC specimens, specimen F-8 jacketed in type 304-SS was bowed and ruptured at the top of the fuel section.

At the date of writing only cursory metallography has been done and this has been performed on only the low burnup specimens. In-pile sintering of the mixed UC and PuC powders has been found to only a small degree. Extensive sintering was found in specimen C-44 which was all PuC powder. The only fuel jacket reaction that has been noted is a thin intermittent layer on the fuel side of the fuel cladding interface in PuC specimen C-44. The layer is, as yet, unidentified but may have resulted from a reaction of free plutonium metal from the hypostoichiometric carbide.

Several pellet and vibratorily compacted mixed carbide fuel elements are now under irradiation under fast reactor conditions in EBR-II. These fuel elements include physically mixed UC-20 wt. % PuC powders, solid solution powders, and single phase pellets. The fuel diameter in these specimens is 0.652 cm (0.257 in.). The jacket materials include type 304-SS, vanadium, V-20Ti, Hastelloy-X, and Nb-1 wt. % Zr. The specimens are being operated from 0.66 to 1.0 kW/cm (20 to 30 kW/ft) at jacket surface temperatures between 575°C and 625°C.

4. Metal Fuels

Three recent developments in uranium-plutonium alloy fuels for fast breeder power reactors should be emphasized: (1) additions of zirconium and titanium have been found to increase the melting temperature of uranium-plutonium alloys from 820–910°C for U–Pu–Fs alloys to ~1100°C for the U–Pu–Zr and U–Pu–Ti alloys, (2) uranium-plutonium-zirconium alloys have been found to be compatible with type 304-SS up to temperatures ranging between 800°C and 850°C, and (3) a burnup of 8.7 at. % (33×10^{20} fiss/cm³) has been demonstrated successfully in fuel pins of a uranium-plutonium-fissium alloy.

Fabrication

A procedure known as injection casting was developed for the fuel rods for EBR-II.⁽⁹⁾ Over 20,000 uranium-fissium alloy fuel pins have been produced for EBR-II using the injection casting procedure. Many of the pins which have been cast more recently have used recycled fuel from the reactor and the entire operation has been done remotely.

A problem which exists in most uranium-plutonium alloys, clad with stainless steel, is the possibility that a temperature excursion can lead to the formation of a molten eutectic between the fuel and clad material. For example, type 304-SS and U-20 wt. % Pu-10 wt. % Mo alloy form a liquid phase at approximately 600°C. Since this is below the operating temperature

of the fuel surface in most fast reactors, it is evident that nickel or iron alloys cannot be used for cladding fuel alloys of this type without a suitable refractory metal barrier or tubing entirely of refractory material. A series of uranium-plutonium-zirconium alloys are being investigated which are compatible with type 304-SS to temperatures above 800°C.

Engineering Properties

Two principal alloy fuels are under study for use in fast breeder reactors. These are the uranium-plutonium-titanium and uranium-plutonium-zirconium alloys. A third alloy system, the uranium-plutonium-fissium alloys, has been investigated extensively in the past and is currently being phased out of the U.S. programme.

The uranium-plutonium-fissium alloys are characterized by relatively low melting temperatures (820–910°C),⁽¹⁰⁾ relatively high irradiation swelling rates,⁽¹¹⁾ and a serious incompatibility with cladding materials based on nickel or iron alloys.⁽¹²⁾ Recent fast reactor metal fuel programmes have been directed towards the development of higher melting U-Pu alloys (~1100°C), primarily alloys of zirconium and titanium,⁽¹³⁾ which are also the least undesirable from physics considerations.

Compatibility diffusion couples between type 304-SS and U-Pu-Fs and U-Pu-Ti alloys showed evidence of melting after 7 days' exposure at 650°C. On the other hand, U-Pu-Zr alloy couples against stainless steel showed essentially no reaction. At 800°C a very thin reaction layer was apparent. At 850°C melting was first observed. It has been found that cladding alloys high in nickel are generally not as compatible with the fuel alloy as are cladding alloys high in iron.

Irradiation Behaviour

The U-Pu-Fs alloys are characterized by their tendency to swell at relatively low temperatures, of the order of 370°C.⁽¹¹⁾ Subsequent experiments have shown that the swelling of U-Pu-Fs alloys can be minimized through the use of clad restraint.⁽¹⁴⁾ The early irradiations on U-Pu-Fs alloys used thin, 0.023 cm (0.009 in.), cladding of refractory metal alloys, which achieved burnups in the range of 2 to 3 at.%, successfully. Recent experiments have evaluated the effect on achievable burnup of cladding thicknesses in the range of 0.0380–0.0635 cm (0.015–0.025 in.). Post-irradiation examination has begun on a group of five such fuel pins that successfully achieved burnups of 7.1–8.7 at.%. Five specimens were intact while a sixth specimen was found to have developed a localized jacket rupture after 8.7 at.% burnup. The specimens were irradiated for 29 months at maximum clad surface temperatures ranging from 545°C to 400°C (end of irradiation).

Irradiations have begun more recently on the high melting U-Pu-Ti and

U-Pu-Zr alloys. A fuel pin specimen of U-15 wt.% Pu-10 wt.% Ti alloy is being irradiated in V-20 wt.% Ti alloy tubing with maximum irradiation temperatures of 760°C in the fuel and 500°C on the cladding surface. A non-destructive examination at 4.6 at.% burnup showed that the element was intact. A fuel pin specimen of U-18.5 wt.% Pu-14.1 wt.% Zr alloy is also being irradiated in V-20 wt.% Ti alloy tubing. In this case, the maximum irradiation temperatures are 870°C in the fuel and 600°C on the cladding surface. Examination of this specimen by neutron radiography at 4.2 at.% showed that the specimen was still intact. Both specimens will continue irradiation to determine the point of jacket failure. EBR-II irradiations, to burnups ranging from 5 to 10 at.% with maximum clad surface temperatures ranging up to 650°C, on the zirconium and titanium alloys are being performed.

5. Cladding

Cladding Materials Selection

There is an increasingly widely held conviction that the principal problem in fast reactor fuels is not in the fuel itself, but in developing an adequate cladding which is stable with respect to the core material, the fission products, irradiation damage within the cladding, and reaction with the coolant. Mechanical as well as chemical interactions must be considered.

In-reactor evaluations of candidate clad materials are proceeding in the EBR-II tests. These include types 304, 316, 347, and 321 SS, Incoloy 800, Hastelloy-X, etc., with various wall thicknesses. Out-of-reactor tests are in progress, as well, to determine the compatibility of these metals with fuel materials and sodium in similar temperature environments. Results of these tests will be forthcoming in the near future.

Irradiation Effects in Stainless Steels at High Temperatures

The effect of irradiation on the stress strain behaviour of structural austenitic stainless steels⁽²⁸⁾ irradiated and tested above 500°C is different than for the alloys irradiated and tested below 500°C. Irradiation of an alloy below 500°C produces a structure that has a higher yield stress, a lower strain hardening coefficient, and hence a lower uniform strain and lower ductility in terms of total elongation than the unirradiated alloy when tested below 500°C. In many instances, no change in true fracture strain or stress is observed for stainless steels irradiated to less than 1×10^{21} neutrons/cm².

Above 500°C the effect of irradiation on these materials is different. First, the ductilities in terms of uniform and fracture strains, as well as total elongation, are reduced and the magnitudes of these reductions are highly strain rate sensitive. Generally, changes in yield stress and in the

stress strain relationship are not observed. Since the ductility is reduced, the ultimate tensile strengths and fracture strengths are reduced. The magnitudes of these reductions are highly sensitive to the work hardening capabilities of the alloy.

There are two significant differences between the irradiation effect below 500°C and that above 500°C. First, the radiation damage that affects the low temperature properties can be removed by post-irradiation heat treatment. Much of the damage is removed *in situ* if the irradiation temperature is above about 350°C. However, post-irradiation annealing of iron- and nickel-base alloys at temperatures between 1050°C and 1300°C does not remove the damage that causes the low ductility at elevated temperatures. The thermal stability of the defect causing the low ductility has been illustrated.^(29,30) A second significant difference is the dependence upon the neutron energy or the energy spectrum of the neutron flux. The low temperature damage is due primarily to the production of vacancies and interstitials by fast neutrons. However, the magnitude of the elevated-temperature embrittlement correlates best with thermal neutrons⁽³¹⁾ for normal iron- and nickel-base alloys.

The (n,α) reaction with ^{10}B appears to be the one capable of producing a defect structure having the thermal stability previously noted. However, helium (alpha particles) can be produced from the reaction of fast neutrons with iron, nickel, nitrogen, and other elements, and the quantities of helium produced will be significant for normal engineering alloys irradiated to a dose above 1×10^{21} neutrons/cm². Embrittlement at elevated temperatures is believed to be primarily caused by the production of helium gas.

The nature of the embrittlement is such that the nucleation and rate of propagation of wedge type grain-boundary cracks are affected. Irradiation studies at the Oak Ridge National Laboratory with type 304-SS indicate that the magnitude of irradiation embrittlement can be reduced by several techniques. These are a reduction in grain size,⁽³²⁾ a suitable pre-irradiation heat treatment that produces a particular type of grain-boundary precipitate, and modification of the alloy composition. These methods are believed to be successful because the quantity of helium at the grain-boundary is reduced and, in most cases, the stress necessary to nucleate and propagate a grain-boundary crack is increased.

Examples of the above techniques are indicated in the tables on p. 625.⁽³³⁾

Irradiation Results of Incoloy 800, Hastelloy-X, and Inconel 625

Incoloy 800, Inconel 625, and Hastelloy-X were irradiated at 538–704°C (1000–1300°F) in two different reactors, GETR and EBR-II. GETR has a fast and thermal flux of equal magnitude. EBR-II has a predominantly fast flux. The experiments thus determined the effect of a fast neutron flux, with

(a) *Influence of Grain Size on Irradiation Embrittlement of SS (304)*

ASTM grain size	Total elongation (%)			
	At 704°C		At 842°C	
	Unirradiated	Irradiated	Unirradiated	Irradiated
3	30	19	23	8
5	39	23	36	11
8-10	58	42	47	16

(Type 304-SS, irradiated at 50°C to 2×10^{20} neutrons/cm² (thermal) and 2×10^{19} neutrons/cm (>1 MeV) and strained at a rate of 0.2%/min.)

(b) *Influence of Pre-irradiation Heat Treatment*

Heat treatment	Tensile test temperature	Total elongation
1 hour at 1036°C	704°C	20.5%
1 hour at 1036°C and 100 hours at 800°C	704°C	30.5%
1 hour at 1036°C	842°C	7.6%
1 hour at 1036°C and 100 hours at 800°C	842°C	13.9%

Type 304-SS, 0.06% C, irradiated at 50°C to 7×10^{20} neutrons/cm² (>1 MeV) and 9×10^{20} neutrons/cm² (thermal) and strained at a rate of 0.2%/min.)

(c) *Influence of Titanium Addition to 18-8 SS*

Titanium (wt. %)	Total elongation (%)		
	650°C	700°C	842°C
0	31.0	31.5	20.0
0.2	40.0	38.0	45.1
0.3	34.7	28.5	35.2
0.4	31.5	26.5	25.0
1.0	30.6	23.5	23.5

(Specimens irradiated to 1.5×10^{19} neutrons/cm² (>1 MeV) and 1×10^{20} (thermal) and tested at strain rate of 2%/min for 1 in. gauge length.)

and without thermal neutrons, on the elevated temperature mechanical behaviour of high nickel austenitic alloys of both the solution strengthened and the precipitation hardening types.

The following results were obtained:

- (1) In tensile tests performed at 1000°F (538°C) and 1300±25°F (704°C) (but not in those at 78°F (25°C)) severe reductions in ductility were observed. The residual elongations at 1300°F (704°C) of the irradiated Inconel 625, Hastelloy-X and Incoloy 800 are as follows: 1–2%, 1–3%, and 7–9%, respectively. A change in fracture mode from ductile transgranular to brittle intergranular for both reactor environments was observed. The embrittlement effect is strain rate dependent with greater embrittlement at low strain rates.
- (2) In the precipitation hardening alloys, precipitation rates were greatly enhanced in both radiation environments. In short time irradiations the precipitation hardening alloys were greatly strengthened by precipitation; in long time irradiations they were weakened by over-aging. Thermal control specimens showed only slight changes in properties.
- (3) Irradiation at 1000°F (538°C) to 1300°F (704°C) resulted in some point defect hardening of the solid solution alloy.
- (4) All effects were evident at exposures as low as 3.5×10^{20} neutrons/cm² and increased in severity to the highest exposure, 2.0×10^{21} neutrons/cm² (>1 MeV).

6. Fast Flux Test Facility Programme

Successful development of economically competitive fast breeder reactor systems requires a long-range programme which is geared to the development and evaluation of long-lived fuels and core construction materials. The success of the thermal reactors has been largely a result of the favourable development of in-core materials, tested in the several available thermal test reactors. The U.S. needs, in addition to the EBR-II and Fermi, a versatile facility which will provide for testing high-risk fuels, defected fuels, and vented fuels, as well as provide advanced conditions for material and cladding irradiations.

To obtain the required versatility, the facility is planned to include the following special features: (a) fast fluxes up to 1×10^{16} neutrons/cm² per second; (b) a wide range of core reactivity control to accommodate a variety of test compositions; (c) ability to accommodate and test fuel failures; (d) capability to instrument fuel and coolant channels; and (e) closed loops.

A number of qualified industrial and laboratory organizations in the fast reactor field are now engaged in an evaluation of various conceptual ideas in an effort to arrive at the best method for achieving FFTF objectives.

Driver Fuel Development

Safety, reliability, and fuel cycle cost economy are important characteristics in the selection of the driver core fuel. The driver fuel must be a fuel best suited to the needs of a test reactor to provide an assured testing environment. Although several fuel types are under consideration, early evaluation has centred about two fuels: (1) a PuO_2 -stainless steel cermet, and (2) a mixed oxide ceramic of UO_2 - PuO_2 . Table 2 summarizes the anticipated operating range for the fuel and core.

TABLE 2. DESIGN AND OPERATING RANGE FOR FFTF DRIVER FUEL*

	PuO_2 -SS cermet	PuO_2 - UO_2 ceramic
<i>Fuel</i>		
Volume % PuO_2	18-28†	20-30†
Cladding	304-SS	304-SS
Fuel diameter— <i>inches</i>	0.19-0.24	0.15-0.23
Average fuel temperature†—°C (°F)	450 (850)	1066 (1950)
Peak hot spot centreline temperature‡—°F	693 (1280)	1760 (3200)
Burnup (av.) % Pu^{239} (18 % $\simeq 10^{21}$ fissions/cm ³)	12-27	15-30
<i>Core</i>		
Volume—litres	600-800	600-800
Power level—MW(th)	300-400	240-500
Peak centreline flux (neutrons/cm ² /s)	$1-1.35 \times 10^{16}$	$0.65-1 \times 10^{16}$
Number of fuel subassemblies	61-91	61-91
Pins per subassembly	166-328	268-328
Core diameter— <i>cm (m)</i>	71-112 (28-44)	61-112 (24-44)
H/D	0.75-1.1	0.65-1.1
Coolant inlet temperature—°C (°F)	260-370 (500-700)	260-370 (500-700)
Core ΔT —°C (°F)	93-204 (200-400)	93-204 (200-400)

* All values are tentative and subject to change based on the design selected.

† Range also due to core radial zoning.

‡ Initial target temperatures based on burnup and safety criteria.

The PuO_2 -stainless steel cermet is considered as a possible driver fuel for initial FFTF conditions because of its expected large thermal expansion, high thermal conductivity, and good coolant compatibility.

Co-extrusion fabrication is being considered for FFTF cermet pins because of simplicity, ease of achieving a metallurgical bond, and rapid production of fuel pins; however, the pellet process offers the advantage of better cladding thickness control and is more adaptable to non-destructive quality control testing. PuO_2 particles 100-150 μ in diameter are estimated to be required to minimize matrix damage during irradiation and yet retain

mechanical strength of the matrix. The optimum PuO_2 particle density has not been selected at this time.

$\text{UO}_2\text{-PuO}_2$ is considered as the other reference fuel for FFTF. Bulk density of the mixed oxide would be in the 85% of theoretical density range in order to accommodate swelling and yet avoid slumping in the pins. The SS-clad pins would have a nominal o.d. of about 0.2 in. (~ 5 mm) and a large fission gas plenum (plenum/fuel volume ratio = 0.8–1.5). The major uncertainty with oxide fuels is reliability of thermal expansion and physical behaviour of the fuel during transients.

Alloy fuels, attractive because of experience, high fissile density, high thermal conductivity, and reliable thermal expansion, are considered as a candidate driver fuel. Other fuels such as carbides and nitrides are being evaluated to ensure that core characteristics could take advantage of future successful fuel developments.

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